

# Exchangeability and the Dirichlet Limit of a Multicolor Pólya Urn

Route-A source-local certificate HCS-C263

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## Abstract

We close the positive-reinforcement classical multicolor Pólya urn from ordered histories to its count law and normalized-mass martingale. The proof is exact for every number of colors and every active positive initial mass. This source probability theorem uses no target arithmetic or spectral data.

## 1 Frozen urn

Let  $a_i > 0$  for  $1 \leq i \leq K$ , let  $c > 0$ , and put  $\alpha_i = a_i/c$  and  $A = \sum_i \alpha_i$ . At each integer time a color is drawn proportionally to its current mass, replaced, and receives additional mass  $c$ . Write  $N_i(n)$  for its count among the first  $n$  draws and use the rising factorial  $(x)^{\overline{r}} = x(x+1) \cdots (x+r-1)$ .

**Theorem 1** (word and count law). *If an ordered word  $w = (w_1, \dots, w_n)$  has multiplicities  $n_i$ , then*

$$\mathbb{P}(w) = \frac{\prod_{i=1}^K (\alpha_i)^{\overline{n_i}}}{(A)^{\overline{n}}}. \quad (1)$$

Consequently

$$\mathbb{P}\{N(n) = (n_1, \dots, n_K)\} = \frac{n!}{\prod_i n_i!} \frac{\prod_i (\alpha_i)^{\overline{n_i}}}{(A)^{\overline{n}}}. \quad (2)$$

*Proof.* Before draw  $t+1$ , the predictive probability of color  $i$  is  $(\alpha_i + N_i(t))/(A+t)$ . Multiplication along a word groups the numerator factors by color and yields (1), independent of order. There are  $n!/\prod_i n_i!$  words with the given counts, proving (2). Summing (2) gives one by the multivariate rising-factorial Vandermonde identity.  $\square$

## 2 Marginals and every factorial moment

Write  $(x)_{[r]} = x(x-1) \cdots (x-r+1)$  for a falling factorial. The distinction from the rising factorial in (1)–(2) is essential.

**Theorem 2** (moment closure). *Every color count is beta-binomial:*

$$\mathbb{P}\{N_i(n) = x\} = \binom{n}{x} \frac{(\alpha_i)^{\overline{x}} (A - \alpha_i)^{\overline{n-x}}}{(A)^{\overline{n}}}.$$

For a multi-index  $r = (r_1, \dots, r_K)$  and  $R = \sum_i r_i$ ,

$$\mathbb{E} \prod_i (N_i(n))_{[r_i]} = (n)_{[R]} \frac{\prod_i (\alpha_i)^{\overline{r_i}}}{(A)^{\overline{R}}}. \quad (3)$$

Hence

$$\begin{aligned} \mathbb{E} N_i &= \frac{n \alpha_i}{A}, \\ \text{Var } N_i &= \frac{n \alpha_i (A - \alpha_i) (A + n)}{A^2 (A + 1)}, \\ \text{Cov}(N_i, N_j) &= -\frac{n \alpha_i \alpha_j (A + n)}{A^2 (A + 1)} \quad (i \neq j). \end{aligned}$$

*Proof.* Sum (2) over the other colors and apply rising-factorial Vandermonde to obtain the marginal. To prove (3), mark an ordered selection of  $R$  draw positions, assigning  $r_i$  positions to color  $i$ . Exchangeability makes the probability of that assignment  $\prod_i (\alpha_i)^{\overline{r_i}} / (A)^{\overline{R}}$ ; there are  $(n)_{[R]}$  ordered marked positions. The first and second choices of  $r$  give the displayed mean and covariance formulas.  $\square$

### 3 Normalized-mass martingale

Define

$$P_i(n) = \frac{\alpha_i + N_i(n)}{A + n}.$$

Conditioning on the first  $n$  draws gives

$$\begin{aligned} \mathbb{E}[P_i(n+1) \mid \mathcal{F}_n] &= P_i(n) \frac{\alpha_i + N_i(n) + 1}{A + n + 1} + (1 - P_i(n)) \frac{\alpha_i + N_i(n)}{A + n + 1} \\ &= P_i(n). \end{aligned}$$

Thus each coordinate is a bounded martingale and the vector converges almost surely. The final revision identifies the limit law and closes the singular boundary faces.

### 4 Certificate and Route-A boundary

The exact implementation enumerates ordered words and independently aggregates them by composition. It compares the predictive recursion with (2), verifies every conditional martingale row, and retains exact rational values.

Total mass grows from  $\sum_i a_i$  to  $\sum_i a_i + cn$ . Therefore this owner has no nontrivial periodic source orbit, primitive repetition clock, or source Artin–Mazur product. It supplies no arithmetic local datum, Euler factor, root number, automorphy statement, target divisor, functional equation, or Hilbert–Pólya operator. Under NO\_BAD\_EULER\_OR\_ROOT\_NUMBER, Route B is disabled.

### References

- [1] F. Eggenberger and G. Pólya, *Über die Statistik verketteter Vorgänge*, *Z. Angew. Math. Mech.* **3** (1923), 279–289, [doi:10.1002/zamm.19230030407](https://doi.org/10.1002/zamm.19230030407).
- [2] D. Blackwell and J. B. MacQueen, *Ferguson distributions via Pólya urn schemes*, *Ann. Statist.* **1** (1973), 353–355, [doi:10.1214/aos/1176342372](https://doi.org/10.1214/aos/1176342372).